

# Project Management Module — Official Documentation

**Module Code:** PROJECT\_MANAGEMENT **Version:** 1.0.0 **Date:** 2026-08-21 **Status:** Production Ready

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## 1. Overview

The Project Management module enables law firms to manage their clients' recurring legal work — portal credentials, renewal deadlines, and team assignments. Each project belongs to one client and can have multiple credentials, renewals, and team members.

### Key Features

- **Project Profiles** — Create and manage projects linked to clients
- **Credential Vault** — Store portal credentials with AES-256-GCM encryption
- **Renewal Tracking** — Recurring schedules (yearly, quarterly, monthly) + one-time deadlines
- **Team Management** — Owner + member assignments per project
- **Client Portal** — Limited view for clients to see their projects and renewal deadlines
- **Dashboard** — Overdue alerts, upcoming deadlines, project statistics

## 2. Architecture

### Module Structure

```
src/main/java/com/lawfirm/erp/modules/projectmanagement/
├─ controller/           # REST controllers (6)
├─ service/              # Business logic (6 interfaces + 7 impls)
├─ repository/          # Data access (6 repos)
├─ entity/              # JPA entities (6)
├─ dto/request/          # Inbound DTOs (9)
├─ dto/response/         # Outbound DTOs (9)
├─ enums/               # Type-safe enums (5)
└─ mapper/              # Entity → DTO mapping (1)
```

### Pattern

All services follow the codebase standard:

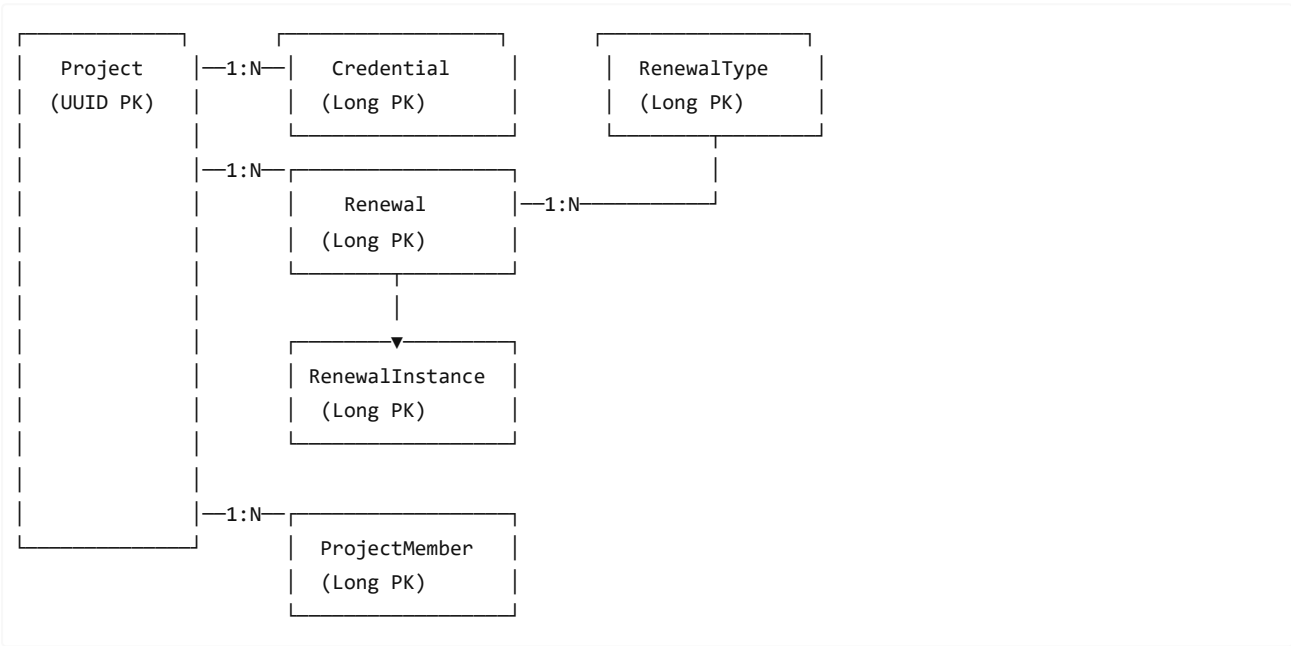
```
Controller → Service (interface) → ServiceImpl → Mapper → Repository
```

### Dependencies

- `ConfigEncryptionUtil` — AES-256-GCM encryption for credentials
- `AuditService` — All mutations are audit-logged
- `CurrentUserResolver` — Resolves the authenticated user
- `FirmContextHolder` — Multi-tenant firm context
- `UserRepository` — User resolution for member names

## 3. Data Model

### Entity Relationship Diagram



### 3.1 Project (Aggregate Root)

| Column                       | Type         | Constraints          | Description                           |
|------------------------------|--------------|----------------------|---------------------------------------|
| <code>id</code>              | UUID         | PK                   | Exposed in URLs and client portal     |
| <code>firm_id</code>         | UUID         | FK → firms, NOT NULL | Multi-tenant scope                    |
| <code>client_user_id</code>  | UUID         | FK → users, NULLABLE | Link to existing client               |
| <code>client_name</code>     | VARCHAR(200) | NOT NULL             | Denormalized client name              |
| <code>project_code</code>    | VARCHAR(30)  | UNIQUE per firm      | Format: FIRMCODE-PRJ-YYYY-NNNNN       |
| <code>name</code>            | VARCHAR(200) | NOT NULL             | Project name                          |
| <code>description</code>     | TEXT         |                      | Project description                   |
| <code>status</code>          | ENUM         | NOT NULL             | ACTIVE, ON_HOLD, COMPLETED, CANCELLED |
| <code>start_date</code>      | DATE         |                      | Project start date                    |
| <code>target_end_date</code> | DATE         | NULLABLE             | End date (null = ongoing)             |
| <code>owner_id</code>        | UUID         | FK → users, NOT NULL | Project manager                       |
| <code>created_at</code>      | TIMESTAMP    |                      | Audit                                 |

|            |           |              |             |
|------------|-----------|--------------|-------------|
| updated_at | TIMESTAMP |              | Audit       |
| created_by | UUID      |              | Audit       |
| is_active  | BOOLEAN   | DEFAULT TRUE | Soft delete |

**Indexes:** (firm\_id, status) , (firm\_id, project\_code) UNIQUE , (client\_user\_id)

### 3.2 Credential

| Column             | Type         | Constraints             | Description                   |
|--------------------|--------------|-------------------------|-------------------------------|
| id                 | BIGINT       | PK, AUTO                |                               |
| project_id         | UUID         | FK → projects, NOT NULL |                               |
| site_name          | VARCHAR(150) | NOT NULL                | e.g. "USPTO Portal"           |
| site_type          | VARCHAR(50)  |                         | Free-form: "Trademark Portal" |
| site_url           | VARCHAR(500) |                         | Redirect URL                  |
| username_or_email  | VARCHAR(200) | NOT NULL                |                               |
| encrypted_password | TEXT         | NOT NULL                | AES-256-GCM encrypted         |
| contact_person     | VARCHAR(150) |                         | Who gave us credentials       |
| contact_phone      | VARCHAR(15)  |                         |                               |
| contact_email      | VARCHAR(200) |                         |                               |
| notes              | TEXT         |                         |                               |
| is_active          | BOOLEAN      | DEFAULT TRUE            | Soft delete                   |

### 3.3 RenewalType

| Column      | Type         | Constraints          | Description                   |
|-------------|--------------|----------------------|-------------------------------|
| id          | BIGINT       | PK, AUTO             |                               |
| firm_id     | UUID         | FK → firms, NULLABLE | NULL = system default         |
| name        | VARCHAR(100) | NOT NULL             | e.g. "Trademark Renewal"      |
| description | VARCHAR(250) |                      |                               |
| is_system   | BOOLEAN      | DEFAULT FALSE        | System types can't be deleted |
| is_active   | BOOLEAN      | DEFAULT TRUE         |                               |

### 3.4 Renewal (Recurrence Template)

| Column          | Type   | Constraints                  | Description |
|-----------------|--------|------------------------------|-------------|
| id              | BIGINT | PK, AUTO                     |             |
| project_id      | UUID   | FK → projects, NOT NULL      |             |
| renewal_type_id | BIGINT | FK → renewal_types, NOT NULL |             |

|                |              |              |                                      |
|----------------|--------------|--------------|--------------------------------------|
| title          | VARCHAR(200) | NOT NULL     |                                      |
| description    | TEXT         |              |                                      |
| recurrence     | ENUM         | NOT NULL     | ONE_TIME, YEARLY, QUARTERLY, MONTHLY |
| start_date     | DATE         | NOT NULL     | First occurrence                     |
| end_date       | DATE         | NULLABLE     | Stop generating after this           |
| assigned_to_id | UUID         | FK → users   | Who handles this                     |
| status         | ENUM         | NOT NULL     | ACTIVE, COMPLETED, CANCELLED         |
| is_active      | BOOLEAN      | DEFAULT TRUE |                                      |

### 3.5 RenewalInstance (Each Deadline)

| Column          | Type      | Constraints             | Description                                       |
|-----------------|-----------|-------------------------|---------------------------------------------------|
| id              | BIGINT    | PK, AUTO                |                                                   |
| renewal_id      | BIGINT    | FK → renewals, NOT NULL |                                                   |
| due_date        | DATE      | NOT NULL                |                                                   |
| status          | ENUM      | NOT NULL                | PENDING, IN_PROGRESS, COMPLETED, OVERDUE, SKIPPED |
| completed_at    | TIMESTAMP | NULLABLE                |                                                   |
| completed_by_id | UUID      | FK → users              |                                                   |
| notes           | TEXT      |                         |                                                   |

### 3.6 ProjectMember

| Column          | Type   | Constraints             | Description           |
|-----------------|--------|-------------------------|-----------------------|
| id              | BIGINT | PK, AUTO                |                       |
| project_id      | UUID   | FK → projects, NOT NULL |                       |
| user_id         | UUID   | FK → users, NOT NULL    |                       |
| role_in_project | ENUM   | NOT NULL                | OWNER, MEMBER, VIEWER |

**Unique constraint:** (project\_id, user\_id)

---

## 4. API Reference

**Base URL:** /api/v1

All endpoints require JWT authentication via `Authorization: Bearer <token>` header.

### 4.1 Projects

#### Create Project

POST /api/v1/projects

**Request Body:**

```
{
  "name": "Nepal Telecom Trademark Renewal",
  "clientName": "Nepal Telecom Ltd.",
  "clientId": "uuid-of-existing-client",
  "description": "Annual trademark renewal for Nepal Telecom",
  "startDate": "2026-01-15",
  "targetEndDate": "2028-12-31",
  "ownerId": "uuid-of-project-manager"
}
```

**Response:** 201 Created

```
{
  "success": true,
  "message": "Project created",
  "data": {
    "id": "uuid",
    "projectCode": "APX-PRJ-2026-00001",
    "name": "Nepal Telecom Trademark Renewal",
    "clientName": "Nepal Telecom Ltd.",
    "status": "ACTIVE",
    "ownerName": "Ram Shrestha",
    "members": [...],
    "credentialCount": 0,
    "renewalCount": 0,
    "overdueInstances": 0
  }
}
```

**List Projects**

```
GET /api/v1/projects?status=ACTIVE&page=0&size=20
```

**Query Parameters:**

| Param  | Type   | Required | Description              |
|--------|--------|----------|--------------------------|
| status | String | No       | Filter by status         |
| search | String | No       | Search by name           |
| page   | int    | No       | Page number (default: 0) |
| size   | int    | No       | Page size (default: 20)  |

**Get Project**

```
GET /api/v1/projects/{projectCode}
```

**Update Project**

```
PUT /api/v1/projects/{projectCode}
```

### Change Status

```
PATCH /api/v1/projects/{projectCode}/status?status=ON_HOLD
```

### Add Team Member

```
POST /api/v1/projects/{projectCode}/members
```

#### Request Body:

```
{
  "userId": "uuid-of-user",
  "role": "MEMBER"
}
```

### Remove Team Member

```
DELETE /api/v1/projects/{projectCode}/members/{userId}
```

### List Team Members

```
GET /api/v1/projects/{projectCode}/members
```

## 4.2 Credentials

### Add Credential

```
POST /api/v1/projects/{projectCode}/credentials
```

#### Request Body:

```
{
  "siteName": "USPTO Portal",
  "siteType": "Trademark Portal",
  "siteUrl": "https://www.uspto.gov",
  "usernameOrEmail": "nepaltelecom@legal.com",
  "password": "actual-password-here",
  "contactPerson": "Sita Sharma",
  "contactPhone": "+977-9841234567",
  "contactEmail": "sita@uspto.gov",
  "notes": "Portal access for trademark filing"
}
```

**Response:** 201 Created

```
{
  "success": true,
  "message": "Credential added",
  "data": {
    "id": 1,
    "siteName": "USPTO Portal",
    "siteType": "Trademark Portal",
    "siteUrl": "https://www.uspto.gov",
    "usernameOrEmail": "nepaltelecom@legal.com",
```

```
    "password": ".....",
    "contactPerson": "Sita Sharma",
    "createdAt": "2026-08-21T10:30:00"
  }
}
```

### List Credentials

```
GET /api/v1/projects/{projectCode}/credentials
```

**Note:** Passwords are always masked as `.....` in list responses.

### Reveal Password

```
POST /api/v1/projects/{projectCode}/credentials/{id}/reveal
```

#### Response:

```
{
  "success": true,
  "message": "Password revealed (audit-logged)",
  "data": {
    "password": "actual-decrypted-password"
  }
}
```

⚠ **Security:** This endpoint is audit-logged. The reveal event (who, when, which credential) is recorded.

## 4.3 Renewals

### Create Renewal

```
POST /api/v1/projects/{projectCode}/renewals
```

#### Request Body:

```
{
  "renewalTypeId": 1,
  "title": "Nepal Telecom Trademark Yearly",
  "description": "Annual trademark renewal filing",
  "recurrence": "YEARLY",
  "startDate": "2026-03-15",
  "endDate": "2029-03-15",
  "assignedToId": "uuid-of-handling-user"
}
```

**Response:** Auto-generates `RenewalInstance` rows based on recurrence.

### List Renewals

```
GET /api/v1/projects/{projectCode}/renewals
```

### Get Renewal + Instances

```
GET /api/v1/projects/{projectCode}/renewals/{id}
```

### Update Instance Status

```
PATCH /api/v1/projects/{projectCode}/renewals/{renewalId}/instances/{instanceId}
```

#### Request Body:

```
{
  "status": "COMPLETED",
  "notes": "Filed on time, confirmation received"
}
```

## 4.4 Renewal Types

### List Types

```
GET /api/v1/projects/renewal-types
```

### Create Custom Type

```
POST /api/v1/projects/renewal-types
```

#### Request Body:

```
{
  "name": "Environmental Compliance",
  "description": "Annual environmental compliance filing"
}
```

### Delete Custom Type

```
DELETE /api/v1/projects/renewal-types/{id}
```

**Note:** System types ( `isSystem=true` ) cannot be deleted.

## 4.5 Client Portal

### List My Projects

```
GET /api/v1/client/projects
```

**Note:** Returns only projects where `clientId = current user` . No credentials or internal notes.

### View Project

```
GET /api/v1/client/projects/{projectCode}
```

### View Renewal Deadlines

```
GET /api/v1/client/projects/{projectCode}/renewals
```

## 4.6 Dashboard



Get Dashboard Stats

```
GET /api/v1/projects/dashboard
```

Response:

```
{
  "totalProjects": 15,
  "activeProjects": 12,
  "overdueInstances": 3,
  "upcomingInstances": 8,
  "totalCredentials": 24,
  "totalRenewals": 45,
  "overdueItems": [
    {
      "projectCode": "APX-PRJ-2026-00001",
      "projectName": "Nepal Telecom Trademark",
      "renewalTitle": "Annual Trademark Filing",
      "renewalTypeName": "Trademark Renewal",
      "dueDate": "2026-08-01",
      "daysOverdue": 20
    }
  ],
  "upcomingItems": [...]
}
```

5. RBAC & Permissions

Module: PROJECT\_MANAGEMENT

| Permission        | Scope    | SUPER_ADMIN | FIRM_ADMIN | ADVOCATE   | PARALEGAL  | CLIENT  |
|-------------------|----------|-------------|------------|------------|------------|---------|
| VIEW              | TENANT   | ✓           | ✓          | ✓          | ✓          | ✓ (own) |
| CREATE            | TENANT   | ✓           | ✓          | ✗          | ✗          | ✗       |
| EDIT              | TENANT   | ✓           | ✓          | ✓ (owned)  | ✗          | ✗       |
| DELETE            | TENANT   | ✓           | ✓          | ✗          | ✗          | ✗       |
| CREDENTIAL_VIEW   | ASSIGNED | ✓           | ✓          | ✓ (member) | ✓ (member) | ✗       |
| CREDENTIAL_REVEAL | ASSIGNED | ✓           | ✓          | ✓ (owner)  | ✗          | ✗       |

Access Rules

- **Project creation:** Only SUPER\_ADMIN and FIRM\_ADMIN
- **Project editing:** SUPER\_ADMIN, FIRM\_ADMIN, and project OWNER
- **Credential access:** Only team members (owner + members)
- **Credential reveal:** Only project OWNER (audit-logged)
- **Client portal:** Only the linked client user can view their projects
- **Non-admin users:** See only projects they're members of

6. Security — Credential Encryption

## Encryption Details

- **Algorithm:** AES-256-GCM (Galois/Counter Mode)
- **Key:** 256-bit, stored in `config.encryption.key` (Base64-encoded)
- **IV:** 12-byte random IV per encryption
- **Tag:** 128-bit authentication tag

## How It Works

1. **On Save:** Password is encrypted using `ConfigEncryptionUtil.encrypt()` and stored as `encrypted_password`
2. **On List/Detail:** Password is returned as `*****` (never decrypted in bulk)
3. **On Reveal:** Password is decrypted using `ConfigEncryptionUtil.decrypt()` — only when explicitly requested
4. **Audit:** Every reveal event is logged with: user ID, credential ID, site name, timestamp

## Reuse

The same `ConfigEncryptionUtil` is used for:

- SMTP email config passwords
- Project management credential passwords

---

## 7. Renewal Engine

### Recurrence Types

| Type      | Behavior                                   |
|-----------|--------------------------------------------|
| ONE_TIME  | Single deadline on <code>start_date</code> |
| YEARLY    | One instance per year                      |
| QUARTERLY | One instance per quarter (3 months)        |
| MONTHLY   | One instance per month                     |

### Instance Generation

When a renewal is created with `recurrence != ONE_TIME` :

1. **With `endDate`** : Generate instances from `startDate` to `endDate`
2. **Without `endDate`** : Generate instances for 3 years ahead from `startDate`

Example: `YEARLY` starting `2026-03-15` with no `endDate` → generates instances for:

- 2026-03-15
- 2027-03-15
- 2028-03-15

### Daily Overdue Check

A `@Scheduled` job runs daily at **08:00 AM**:

Marks `PENDING` instances with `dueDate < today` as `OVERDUE`

This is implemented in `RenewalScheduler.java` .

### Instance Status Flow

`PENDING` → `IN_PROGRESS` → `COMPLETED`

↓

OVERDUE (auto-detected)



SKIPPED (manual)

## 8. Client Portal

### Access

- Uses existing `CLIENT` user type
- Client must be linked to a project via `clientId`
- Client can only see projects where `clientId = current user`

### What Clients See

| Field                       | Visible | Notes                   |
|-----------------------------|---------|-------------------------|
| Project name, status, dates | ✓       |                         |
| Project description         | ✓       |                         |
| Renewal deadlines           | ✓       | PENDING + OVERDUE only  |
| Renewal status              | ✓       |                         |
| Credentials                 | ✗       | Never exposed to client |
| Team members                | ✗       | Internal only           |
| Internal notes              | ✗       |                         |

### API Endpoints

```
GET /api/v1/client/projects
GET /api/v1/client/projects/{projectCode}
GET /api/v1/client/projects/{projectCode}/renewals
```

## 9. Restrictions & Business Rules

### Project Rules

1. **One client per project** — `clientId` is a single UUID, not a list
2. **Project code is immutable** — Generated at creation, never changes
3. **Owner cannot be removed** — The project owner cannot be removed from members
4. **Owner is auto-added** — When a project is created, the owner is automatically added as a `OWNER` member
5. **Status transitions** — Any status can transition to any other status (no restricted transitions for V1)

### Credential Rules

1. **Passwords are never stored in plaintext** — Always encrypted with AES-256-GCM
2. **Passwords are never returned in list/detail** — Always masked as `••••••`
3. **Password reveal is audit-logged** — Every reveal event is recorded
4. **Soft delete** — Credentials are deactivated, not hard-deleted

### Renewal Rules

1. **Recurrence auto-generates instances** — YEARLY generates 3 years ahead by default
2. **One-time renewals get exactly 1 instance**
3. **Overdue is auto-detected** — Daily scheduler marks past-due PENDING instances as OVERDUE

4. **System renewal types cannot be deleted** — Only firm-scoped custom types can be removed

## Member Rules

1. **Unique membership** — A user can only be a member of a project once (unique constraint)
  2. **Owner protection** — The project owner cannot be removed
  3. **Firm-scoped** — Only users belonging to the same firm can be added as members
- 

# 10. Frontend Integration Guide

## Project List Page

```
GET /api/v1/projects?page=0&size=20&status=ACTIVE
```

- Display: project code, name, client name, status, owner, credential count, renewal count, overdue badge
- Filter: by status (dropdown)
- Sort: by createdAt DESC (default)

## Project Detail Page

```
GET /api/v1/projects/{projectCode}
```

### Tabs:

1. **Overview** — Name, description, status, dates, owner
2. **Credentials** — List with masked passwords, "Reveal" button, "Add" form
3. **Renewals** — List with instances, status badges, "Add" form
4. **Team** — Member list with roles, "Add Member" / "Remove" actions

## Credential Card

```
<CredentialCard
  siteName={credential.siteName}
  siteType={credential.siteType}
  siteUrl={credential.siteUrl}      // "Open Portal" link
  username={credential.usernameOrEmail}
  password={credential.password}    // "....." until revealed
  onReveal={() => POST /credentials/{id}/reveal}
  contact={credential.contactPerson}
/>
```

## Renewal Timeline

```
<RenewalTimeline
  renewals={renewals.map(r => ({
    title: r.title,
    type: r.renewalTypeName,
    instances: r.instances.map(i => ({
      dueDate: i.dueDate,
      status: i.status,           // PENDING, OVERDUE, COMPLETED
      daysOverdue: i.status === 'OVERDUE' ? calculateDays(i.dueDate) : null
    }))
  ))}
/>
```

### Dashboard Cards

```
<DashboardStats
  totalProjects={data.totalProjects}
  activeProjects={data.activeProjects}
  overdueCount={data.overdueInstances}
  upcomingCount={data.upcomingInstances}
/>

<OverdueAlerts items={data.overdueItems} />
<UpcomingDeadlines items={data.upcomingItems} />
```

### Client Portal

```
// Only for CLIENT role users
<ClientProjectList
  endpoint="/api/v1/client/projects"
  // Shows: name, status, dates, upcoming renewals
  // Hides: credentials, team, internal notes
/>
```

### Form Validation

| Field                  | Required | Validation                                 |
|------------------------|----------|--------------------------------------------|
| Project name           | Yes      | Max 200 chars                              |
| Client name            | Yes      | Max 200 chars                              |
| Client user ID         | No       | Must exist in firm                         |
| Start date             | No       | Date                                       |
| Target end date        | No       | Date, must be after start                  |
| Site name (credential) | Yes      | Max 150 chars                              |
| Username (credential)  | Yes      | Max 200 chars                              |
| Password (credential)  | Yes      | Min 1 char (encrypted on save)             |
| Renewal title          | Yes      | Max 200 chars                              |
| Renewal type ID        | Yes      | Must exist                                 |
| Recurrence             | Yes      | Enum: ONE_TIME, YEARLY, QUARTERLY, MONTHLY |
| Start date (renewal)   | Yes      | Date                                       |

---

## 11. Seed Data

### Auto-Seeded on Application Start

#### Module & Permissions

Module: PROJECT\_MANAGEMENT

Permissions:

- PROJECT\_MANAGEMENT:ACCESS
- PROJECT\_MANAGEMENT:VIEW
- PROJECT\_MANAGEMENT:CREATE
- PROJECT\_MANAGEMENT:EDIT
- PROJECT\_MANAGEMENT:DELETE
- PROJECT\_MANAGEMENT:CREDENTIAL\_VIEW
- PROJECT\_MANAGEMENT:CREDENTIAL\_REVEAL

#### Default Renewal Types (System)

| Name                   | Description                              |
|------------------------|------------------------------------------|
| Trademark Renewal      | Annual trademark renewal and maintenance |
| Patent Renewal         | Patent maintenance and renewal fees      |
| License Renewal        | Business license and permit renewals     |
| Annual Compliance      | Annual statutory compliance filings      |
| Tax Filing             | Tax return and filing deadlines          |
| Secretarial Compliance | Company secretarial compliance filings   |
| Other                  | General renewal or deadline              |

## 12. Testing

### Unit Tests

```
# Run all tests
./mvnw test

# Run only project management tests
./mvnw test -Dtest='ProjectServiceTest,RenewalServiceTest'
```

### Test Coverage

| Class              | Tests    | Coverage                                          |
|--------------------|----------|---------------------------------------------------|
| ProjectServiceTest | 5        | Create, validation, members, owner protection     |
| RenewalServiceTest | 4        | One-time, yearly, quarterly recurrence, not-found |
| <b>Total</b>       | <b>9</b> |                                                   |

### Pre-existing Tests

All 132 original tests continue to pass (138 total with new module tests).

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